OpenBench — Research & Audit

Hypothesis verdicts · sources · loop log · judge-style scorecard

Team Inferno · prepared 27 October 2026. Case figures are fixed; research is used only as context and never replaces case data.

1 · Research log and hypothesis verdicts

ID	Hypothesis	Verdict	Evidence
H1	Listing alone fails on reliability	Supported	I-STEM executed 44,084 of 88,887 booking requests [S1]; users settle payment directly with labs [S3]; OpenBench pilot: 38/86 bookable, 14% cancellations (Ex.1).
H2	Direct lab incentives raise sharing	Supported (policy), not yet tested in data	SRIMAN: sharing promoted through incentives and rating systems [S2]; SATHI centres built for startups and MSMEs [S8]. No outcome data found.
H3	Customers pay for guarantees; Cluster B won't pay 28%	Mixed (premium now evidenced)	Auriga Research (NABL) charges a 50% express surcharge [S9]; managed marketplaces justify higher take rates through curation and risk-bearing [S14]; Science Exchange CEO: too-high B2B take rates become untenable [S4]. No India price-elasticity data found.
H4	Flat lab subscriptions adopt slowly in public institutions	Supported	Only 4 of 10 lab managers would consider one (Ex.3); Stanford rejected a 5% fee partly because it fell on grant-funded users [S6].
H5	Committed slots + rerouting cut cancellations more than software	Not verified	Logic plus Exhibit 3 (managed 94% vs SaaS 86%); no external study found. Stated as team assumption.
H6	Tranching tied to quality KPIs is acceptable to investors	Not verified	No India-specific source found in this pass. Presented as a proposal, not a claim.

2 · Sources (accessed October 2026)

ID	Source	Used for
S1	Office of the PSA, I-STEM national footprint (31 Dec 2025) — https://www.psa.gov.in/CMS/web/i-stem	≈50% execution rate (slides 2, 8)
S2	DST SRIMAN Guidelines highlight — https://www.indiascienceandtechnology.gov.in/stihighlights/enabling-availability-and-accessibility-scientific-infrastructure-through-sriman	Lab incentives (slide 4)
S3	I-STEM Quick Reference Guide for Users — https://www.istem.gov.in/uploads/banner/Quick_Reference_Guide_for_Users.pdf	Portal vs Assured (slide 8)
S4	Bowery Capital interview, Science Exchange CEO — https://bowerycap.com/blog/insights/from-the-front-lines-elizabeth-iorns-science-exchange/	Take-rate ceiling (slide 4)
S5	NABL accreditation under ISO/IEC 17025 — https://www.slideshare.net/slideshow/nablpptx-253756935/253756935	Calibration flag (slide 8)
S6	Stanford Daily on core facilities and Science Exchange — https://stanforddaily.com/?p=1073570	Institutional refusal risk (slide 10)
S7	Platform Chronicles, 'Platform leakage' — https://platformchronicles.substack.com/p/platform-leakage	Disintermediation (slide 10)
S8	DST SATHI centres — https://dst.gov.in/centres-housing-major-analytical-instruments-set-increase-access-research-testing-facilities	Anchor-host idea (Q&A)

ID	Source	Used for
S9	Auriga Research, testing charges — https://aurigaresearch.com/testing-charges/	50% express surcharge → A23 priority premium (slide 7, A2)
S10	upGrowth, CAC benchmarks for Indian B2B SaaS by ARR band (2026) — https://upgrowth.in/cac-benchmarks-indian-b2b-saas-arr-band-2026/	CAC context for A31 (slide 6, A2)
S11	Atal Innovation Mission (72 Atal Incubation Centres) — https://en.wikipedia.org/wiki/Atal_Innovation_Mission	Lead pool and free channel (A7, A33)
S12	IMPRI, SATHI 2022-23: 5 centres, 32,000+ users, ~25% MSMEs/startups — https://www.impriindia.com/insights/sathi-india-2022-23/	Lead pool and free channel (A7, A33)
S13	MMH, research on supplier performance and customer loyalty (Grant Thornton data) — https://www.mmh.com/article/research_examines_link_between_supplier_performance_and_customer_loyalty_wh	A27 sensitivity range (A5, slide 8)
S14	TechCrunch, 'Anatomy of a managed marketplace' — https://techcrunch.com/2017/05/25/anatomy-of-a-managed-marketplace/	H3 (research log only; was S9)

3 · Change log (research-update loop)

Iter.	What changed	Why	Score
1	Built model (3 scenarios, 30 inputs), deck, presenter pack. Found base case never broke even with fixed costs of ₹6.8L; switched to a lean city cost (₹0.5L) and capture anchored on the pilot (85%).	First run: M18 EBITDA −₹3.1L/month at 75% capture; 75% had no anchor in the case data.	74
1	Fixed bug: pilot monthly contribution formula multiplied revenue instead of contribution.	Caught when Unit Economics showed a ₹2.53L loss vs the correct ₹2.95L.	74
1	Added reliability ramp: on-time rises from 78% to tier level over 6 months.	Model first showed 94% on-time in Month 1, which no judge would believe.	74
2	Added lab income comparison (pilot lab +3.5% under Assured) to model and slide 4.	Red-team: 'labs keep 72% instead of 82%, why join?' was unanswered.	77
2	Re-opened 5 sources (S1, S2, S4, S6, S7): all claims confirmed. Added the Stanford 15% external-use cap to the Q&A bank.	Spot-check rule; new risk found in S6.	77
2	Disclosed lever dependence on slide 6 and in Sensitivity tab (both levers off: M18 EBITDA −₹0.81L).	Honesty: break-even depends on two team assumptions.	77
2	Fixed tornado labels and chevron text clipping found in visual QA.	Rendered slides checked one by one.	77
3	Design upgrade to the 'Precision Lab' theme: Lab Navy + one Signal Cyan accent, Georgia titles + Arial body, section tracker, 'So what' callouts, calibration-tick footer.	Design prompt (consulting-deck research: action titles, 2 fonts, 3–4 colours, direct labels).	Design 78
3	New exhibits: issue tree (slide 3), native waterfall (slide 4), heat-shaded scoring matrix (slide 5), Gantt roadmap (slide 6), 2×2 risk matrix (slide 11); tornado moved to appendix A5.	Brief asked for the simplest chart that proves each claim.	Design 78
4	Raised all body text to ≥16 pt; turned two dense text blocks into tables; chart labels to ≥12 pt; restored dropped figures (downside column, repeat milestones, 8% late).	Design audit: legibility cap (body under 16 pt) and number-lock diff.	Design 88
5	Theme switched to 'Lab Notebook': graph-paper background, Fountain Ink + Lab Teal + Marker Yellow highlighter, lab-report tracker (Abstract → Controls), sample-tag RESULT boxes, report-header cover with CALIBRATED stamp, closing tagline 'Booked should mean delivered.'	Theme prompt: a theme that tells OpenBench's story.	Theme 87
5	Lab Teal deepened 00897B → 00796B and Red Pen D64545 → C62828 so every text pair passes WCAG 4.5:1; cards turned white with a grid-line border so they stand off the paper.	Contrast script (16 pairs) and visual QA.	Theme 87
6	Team name changed to Team Inferno (Mahi · Minakshi Singha · Swapnil Behera) in every file and script line.	User instruction.	—
7	Added customer-acquisition cost: A31 ₹1,500 per paid-acquired org, A32 0.405 orgs per booking (170/420, Ex.1), A33 80% won through free channels. New monthly cost line in all scenarios.	Gap 1: customers do not simply arrive in new cities.	Feasibility +2

Iter.	What changed	Why	Score
7	CAC alone pushed base M18 EBITDA to −₹0.46L (no break-even). Raised A23 Priority premium 30% → 50% on evidence (Auriga's published 50% express surcharge [S9]) and disclosed A33 as a partnership lever. Base break-even returns at M15.	Never hide a cost; fix only with cited or disclosed levers. Note: IIT Guwahati, BHU and Roorkee rate cards list no urgent fee, so uptake stays at 15% and is tracked.	—
7	Added lab onboarding funnel (Funnel tab, appendix A7) and a slow-onboarding check (12 new labs: no break-even by M18, cash floor ₹81.0L).	Gap 2.	Feasibility +1 (shared)
7	Added segment view of churn as labelled inference (slide 3 callout, appendix A8).	Gap 3; the case has no repeat split, so none was invented.	Problem +1
7	Added 'one backbone, three price points; only Assured until the M7 gate' (slide 5).	Gap 4.	Strategy +1
7	A27 sensitivity 0.5×/1.0×/1.5× (Repeat check tab, A5) with retention evidence [S13]; 55% repeat counter-offer now reasoned on slide 8.	Gap 6.	Analysis +1
7	Fourth scenario 'Downside + response' (Monthly_DownResponse, A9; slide 7 line).	Gap 7.	Analysis +1
7	Bookings target re-checked: no cited fix closes 2,316 → 2,400, so the 2,200 / 55% counter-offer stays.	Gap 8: never tune to hit a target.	—
7	Fixed stale citation in Excel A11 ([S9] → [S6]) after S9 was reassigned; TechCrunch source renumbered S14.	Cross-file citation check.	—

4 · Earlier audit (before the gap fixes; numbers as they were then)

Method: every criterion starts at 50% of its maximum and rises only with cited evidence; caps from the prompt apply. No credit for length.

Criterion	Max	Score	Evidence	Biggest gap
Problem identification & insight	15	12	3 root causes quantified (44% bookable, 14% cancels, 31%/27% interviews); I-STEM ≈50% proof that the problem is structural (slides 2–3)	No customer-segment view of who churns
Feasibility & implementation plan	20	15	Host-by-host onboarding plan, M7 gate, triggers with numbers, lab income +3.5%, downside cash floor ₹40.9L (slides 5, 7, 9, 10)	Customer acquisition is not modelled
Strategic recommendations & creativity	20	16	Segmented hybrid with Connect crossover (26.7 bookings/host), priority SLA shared with labs, quantified tranche counter-offer (slides 4, 5, 7)	Hybrid adds operating complexity across 3 tiers
Quality of analysis	20	15	Live Excel engine, 3 scenarios, tornado, Python and Excel engines agree; lever dependence disclosed (slide 6, appendix)	Break-even rests on 2 team levers; repeat elasticity has one data point
Presentation, delivery & Q&A readiness	25	19	10 action-titled slides, script 9.3 min, 17-question Q&A bank, cheat sheet	Not yet rehearsed; presenter names are placeholders
TOTAL	100	77	Before fixes: 74	

Gap-fix audit (out of 75; presentation not scored)

Criterion	Max	Before	After	Evidence
Problem identification	15	12	13	Slide 3 'Who leaves' callout + A8 table mapping Ex.1 reasons to Ex.2 clusters, clearly labelled inference. Still no measured split (the case has none).
Feasibility	20	15	17	CAC modelled (A31–A33, Excel EBITDA line, slide 6); funnel 267→16 with cited lead pools [S1][S11][S12] (A7); slow-onboarding check. 80% free share and funnel conversions remain team assumptions.
Strategy	20	16	17	One backbone / Assured-only until M7 gate (slide 5); sequencing matches the model (Connect from M9).
Analysis	20	15	17	Priority premium evidenced [S9]; A27 range with evidence [S13]; 4 scenarios; corrected A5 tornado; Excel 0 errors and matches Python. Break-even still leans on 15% uptake and the 11% learning cut (team assumptions).
TOTAL	75	58	64	No caps triggered: all outside numbers cited, headline numbers identical across files, 10 content slides, theme unchanged, no original wording lost.

Issue status

#	Issue	Status	Evidence
1	CAC	Closed	A31–A33 in Excel and A2; cost line in all 4 scenarios; slide 6 line; base still breaks even at M15.
2	16 new labs ambitious	Closed	Funnel tab and A7; slow check (12 labs) shows the cost honestly: no M18 break-even, cash floor ₹81.0L.
3	No segment view	Partly closed	Inference only (slide 3, A8); the case gives no repeat split. Tracking by cluster from M1 is promised.
4	Three tiers complex	Closed	Slide 5 backbone line; model runs Assured only until M8, Connect from M9.
5	Break-even on two levers	Partly closed	Premium cited [S9] and raised to 50%; 15% uptake and the 11% cost cut are still team assumptions; both-off result now −₹1.23L, disclosed on slide 7.
6	A27 one data point	Closed	0.5–1.5× sensitivity (A5, Repeat check tab); evidence 0.55–0.83× [S13] sits below our 1.0×, which is why we offer 55%.
7	No recovery scenario	Closed	Downside + response: −₹2.32L at M18, floor ₹47.7L (slide 7, A9). It does not rescue M18; stated.
8	Missed targets	Open (by design)	No honest change reaches 2,400 or 60%; counter-offer 2,200 / 55% kept with stronger reasoning (slide 8).

Numbers changed by the fixes (old → new, updated in Excel, deck, script, cheat sheet and Q&A)

Number	Old	New	Why
M18 EBITDA, base	+₹0.19L	+₹0.15L	CAC line added; Priority premium 30% → 50% [S9]
M18 EBITDA, downside	−₹2.61L	−₹2.65L	Same
M18 EBITDA, upside	+₹0.93L	+₹0.88L	Same
Cash used M1–M18, base	₹39.5L	₹38.6L	Same
Cash used M1–M18, downside	₹79.1L	₹78.6L	Same
Upside cash used	₹31.0L	₹30.0L	Same
Lowest cash, downside	₹40.9L	₹41.4L	Same

Number	Old	New	Why
Cash used by M7, downside	₹37.7L	₹37.6L	Same
Both levers off, M18	−₹0.81L	−₹1.23L	CAC now in the cost base
Contribution per booking, M18	₹1,267	₹1,354	Higher priority premium
Pilot lab income under Assured	₹1,26,225 (+3.5%)	₹1,28,775 (+5.6%)	Lab keeps half of a larger premium
Priority SLA price	+30%	+50%	Evidence [S9]
Slide 7 drivers (low / high)	−0.76/+1.14 · −0.56/+0.93 · −0.64/+0.74	−0.84/+1.14 · −0.58/+0.88 · −0.67/+0.70	Re-run
A5 tornado (Δ vs base)	±0.95 · ±0.75 · ±0.59 (priority)	±0.99 · ±0.73 · ±0.98 (priority); new CAC row −0.43/+0.21	Re-run; fixed-cost and Assured-cost rows unchanged
A1 P&L rows	—	Revenue, contribution, EBITDA, cash re-read from Monthly_Base ; new 'Acquisition' row	Same
Unchanged	—	M15 · 2,316 · 17 hosts · 92.3% · 57.8% · 387	Checked in all files

Constraint check (C1–C10)

#	Constraint	Result	Evidence
C1	≤10 content slides	PASS	Slides 2–11 are content; slide 1 is the title; 12–20 are appendix (A1–A9)
C2	10-minute talk	PASS	Script is 1,424 words ≈ 9.5 minutes at 150 wpm
C3	Asset-light	PASS	No instrument or lab purchase anywhere; onboarding spend only
C4	Listed ≠ available	PASS	Committed staffed slot blocks (slide 8); bookable-share KPI 44% → ≥80% (slide 9)
C5	Use supplied numbers; label assumptions	PASS	Assumptions A1–A36 in Excel; A2 lists the material ones; exhibits cited on each slide
C6	Projections not guaranteed	PASS	85% capture (65% downside) and a 4-month ramp on every new host
C7	Answer Q1–Q6	PASS	Q1 slides 2–3 · Q2 slide 4 · Q3 slide 5 · Q4 slide 6 · Q5 slide 8 · Q6 slides 9–10
C8	Quantified investor stance	PASS	Slide 7: ₹60L + ₹60L, 2,400→2,200, 60%→55%, covenants
C9	External facts cited	PASS	S1–S14 with URLs; S1 I-STEM 3,812 institutes re-opened and confirmed
C10	Case data beats research	PASS	Research used only as context (I-STEM, SRIMAN, analogues)

Numeric and cross-file checks

• Two engines agree: Python (final2.py) and Excel formulas give the same outputs in all 4 scenarios (base break-even M15, 2,316 bookings, M18 EBITDA ₹14,979 Excel vs ₹15,004 Python, a rounding difference in the CAC

line below ₹100).
• Excel recalculated in LibreOffice: 11,013 formulas, 0 errors.
• Cross-file check of the 10 headline numbers (M15, 2,316, 17, 92.3%, 57.8%, +₹0.15L, −₹2.65L, ₹38.6L, ₹78.6L, 387): identical in the deck, Excel, script, cheat sheet and Q&A.
• Content check: every original line of the 17 slides is still present word for word, except the numbers in the change log.
• Design check: all 20 slides rendered; new elements reuse existing card, table, label and RESULT styles; no overflow or overlap.
• Bug found and fixed in QA: the A5 tornado first showed absolute M18 EBITDA instead of the change vs base; corrected.

Hardest judge questions (red team)

Question	Can the deck answer it?
Your break-even depends on two levers you assumed.	Better. The premium matches a published 50% express surcharge [S9]; uptake (15%) and the 11% cost cut remain assumptions; both off: −₹1.23L (slide 7).
Where does a 1.0 repeat elasticity come from?	Yes. One case data point, tested at 0.5–1.5×; outside evidence 0.55–0.83× [S13] is why we ask for 55% (slide 8, A5).
Can you really onboard 16 new labs in 14 months?	Yes, with a cost. Funnel 267 → 16 (A7); with only 12 we miss M18 break-even but keep ₹81L cash.
What does a customer cost to acquire?	Yes. ₹1,500 per paid-acquired org, 80% free (slide 6, A2); ₹4.6L over 18 months.
What if the downside hits?	Yes. A9: response fired at M9 narrows the loss to −₹2.32L and lifts the cash floor to ₹47.7L; M18 is still missed.

Top fixes still open (ranked by points gained per hour)

#	Fix	Expected gain
1	Rehearse twice to 9:30 with a stopwatch	+2 (presentation)
2	Get 2–3 incubator/SATHI letters of intent to evidence the 80% free channel	+1 (feasibility)
3	Ask 2 pilot labs for their urgent-job share to evidence 15% priority uptake	+1 (analysis)
4	Start logging repeat by cluster now to turn A8 into data	+1 (problem)

Design audit (Precision Lab redesign)

Criterion	Max	Score	Evidence	Gap
Storyline readable from titles alone	15	13	All 10 titles are sentences of ≤15 words and ≤2 lines	Most titles wrap to a second line
Consistency (grid, title position, fonts, colours)	20	18	Same title box, tracker, footer and 0.3-inch spacing on every slide	Georgia shows in a substitute font in our preview; check in PowerPoint
Exhibit quality	20	17	Waterfall, issue tree, heat matrix, Gantt, 2×2 risk map; direct end-labels on the line chart	No break-even marker dot on the line chart
Legibility when projected	15	13	Body ≥16 pt, tables ≥12 pt, chart labels ≥12 pt	Tracker and source lines are 9 pt (captions)
Theme and motif with restraint	10	9	One accent; amber at most once per slide; red only for risk	—
Number integrity	10	10	Number-lock diff vs the previous deck: no number changed; additions read from the Excel model (₹40.9L, sensitivity values)	—

Criterion	Max	Score	Evidence	Gap
PDFs match theme; hand-over ready	10	8	PDFs restyled; PPTX + PDF delivered	Embed fonts in PowerPoint yourself before the event
TOTAL	100	88	Loop 1: 78, Loop 2: 88. Verdict: READY (85 or above)	

Theme audit (Lab Notebook)

Criterion	Max	Score	Evidence	Gap
Theme tells a story that fits OpenBench	15	13	Deck reads as a lab report: tracker Abstract → Controls, RESULT sample tags, EXPERIMENT 04 cover, CALIBRATED stamp	—
Palette discipline	15	13	One accent (teal); highlighter on 3 slides, once each; red only for losses, misses, triggers	Teal and red darkened from the spec for contrast
Contrast and colour-blind safety	15	14	16 of 16 text pairs pass WCAG (lowest 4.56:1); ✓/✗ icons beside every colour judgement	—
Legibility when projected	15	13	Body ≥16 pt; grid texture faint; every chart sits on a solid white panel	Tracker and footer are 9 pt captions
Consistency across deck and PDFs	15	13	Same palette, fonts and 'Lab Book' footer in both PDFs	PDF covers use an ink band, not graph paper
Creative touches with restraint	15	11	Stamp, tags, highlighter, tagline, calibration ticks	Skipped per-section hero icons and the before/after strip to protect space
Number and content integrity	10	10	Number-lock diff: no number changed; only the label 'Experiment 04' added	—
TOTAL	100	87	Verdict: READY (85 or above)	

Contrast check (WCAG)

Pair	Colours	Ratio	Needs	Result
Ink text on paper	1F2A44 / FAF8F3	13.43	≥4.5:1	PASS
Ink on white panel	1F2A44 / FFFFFF	14.26	≥4.5:1	PASS
Ink on zebra row	1F2A44 / F6F7F9	13.3	≥4.5:1	PASS
Ink on highlighter	1F2A44 / FFD54F	10.1	≥3:1	PASS
Teal text on paper	00796B / FAF8F3	5.01	≥4.5:1	PASS
Teal text on white	00796B / FFFFFF	5.32	≥4.5:1	PASS
Teal text on zebra	00796B / F6F7F9	4.96	≥4.5:1	PASS
Red text on white	C62828 / FFFFFF	5.62	≥4.5:1	PASS
Red text on zebra	C62828 / F6F7F9	5.24	≥4.5:1	PASS
Grey caption on paper	6B7280 / FAF8F3	4.56	≥4.5:1	PASS
Grey on white	6B7280 / FFFFFF	4.83	≥4.5:1	PASS
White on ink panel	FFFFFF / 1F2A44	14.26	≥4.5:1	PASS
White on teal (score 82, lanes)	FFFFFF / 00796B	5.32	≥4.5:1	PASS
White on red risk dots	FFFFFF / C62828	5.62	≥4.5:1	PASS

Pair	Colours	Ratio	Needs	Result
Ink on heat tint 5	1F2A44 / 66AFA6	5.6	≥4.5:1	PASS
Ink on heat tint 4	1F2A44 / 99C9C4	7.81	≥4.5:1	PASS

Verdict

Gap-fix audit: 64/75 (was 58). Six of eight issues closed, two partly closed, and the missed investor targets stay open by design because no honest, cited change closes them. Target 66 not reached: the remaining points need evidence only the team can collect (free-channel letters, priority uptake). No caps triggered. Theme and design unchanged; 10 content slides; every original line kept. Earlier design audits: Precision Lab 88/100; Lab Notebook 87/100.